

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	30 June 2026
Team ID	
Project Name	Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

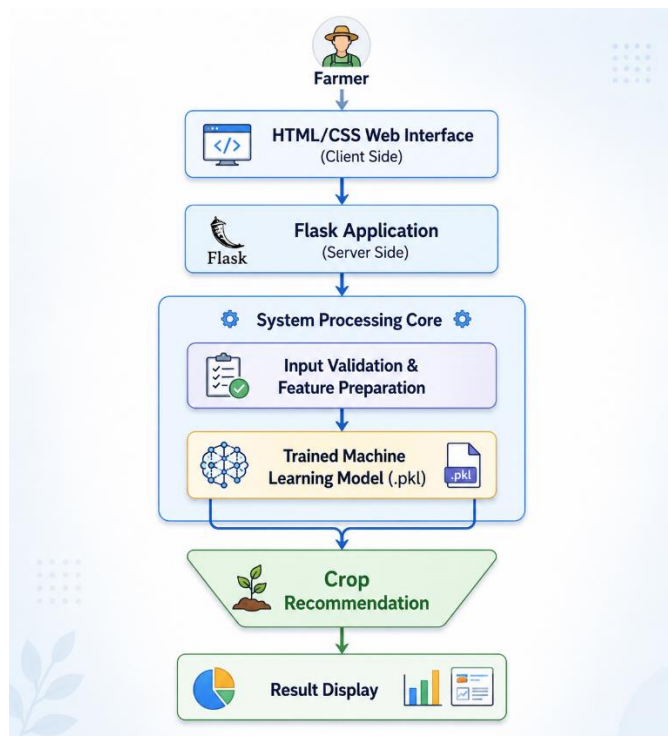


Table-1 : Components & Technologies:

Component	Description	Technology
User Interface	Farmer Web Application	HTML5, CSS3
Backend	Business Logic & Request Handling	Flask (Python)
Machine Learning	Crop Recommendation	Scikit-learn
Data Processing	Data Preparation & Analysis	Pandas, NumPy
Model Loading	Load Trained Prediction Model	Joblib
Development Environment	Model Training & Testing	Jupyter Notebook
Deployment	Local Web Application	Flask Development Server

Table-2: Application Characteristics:

Characteristic	Description	Technology
Open Source Frameworks	Backend & Web Application	Flask, Scikit-learn
User Interface	Responsive Web Interface	HTML5, CSS3
Data Processing	Data Preparation & Analysis	Pandas, NumPy
Machine Learning	Crop Prediction	Scikit-learn, Joblib
Performance	Fast Prediction Response	Flask, Scikit-learn

Table-3: Technology Stack:

Category	Technology
Programming Language	Python

Backend Framework	Flask
Machine Learning	Scikit-learn
Data Processing	Pandas
Numerical Computing	NumPy
Model Storage	Joblib
Frontend	HTML, CSS
IDE	VS Code / PyCharm
Version Control	Git